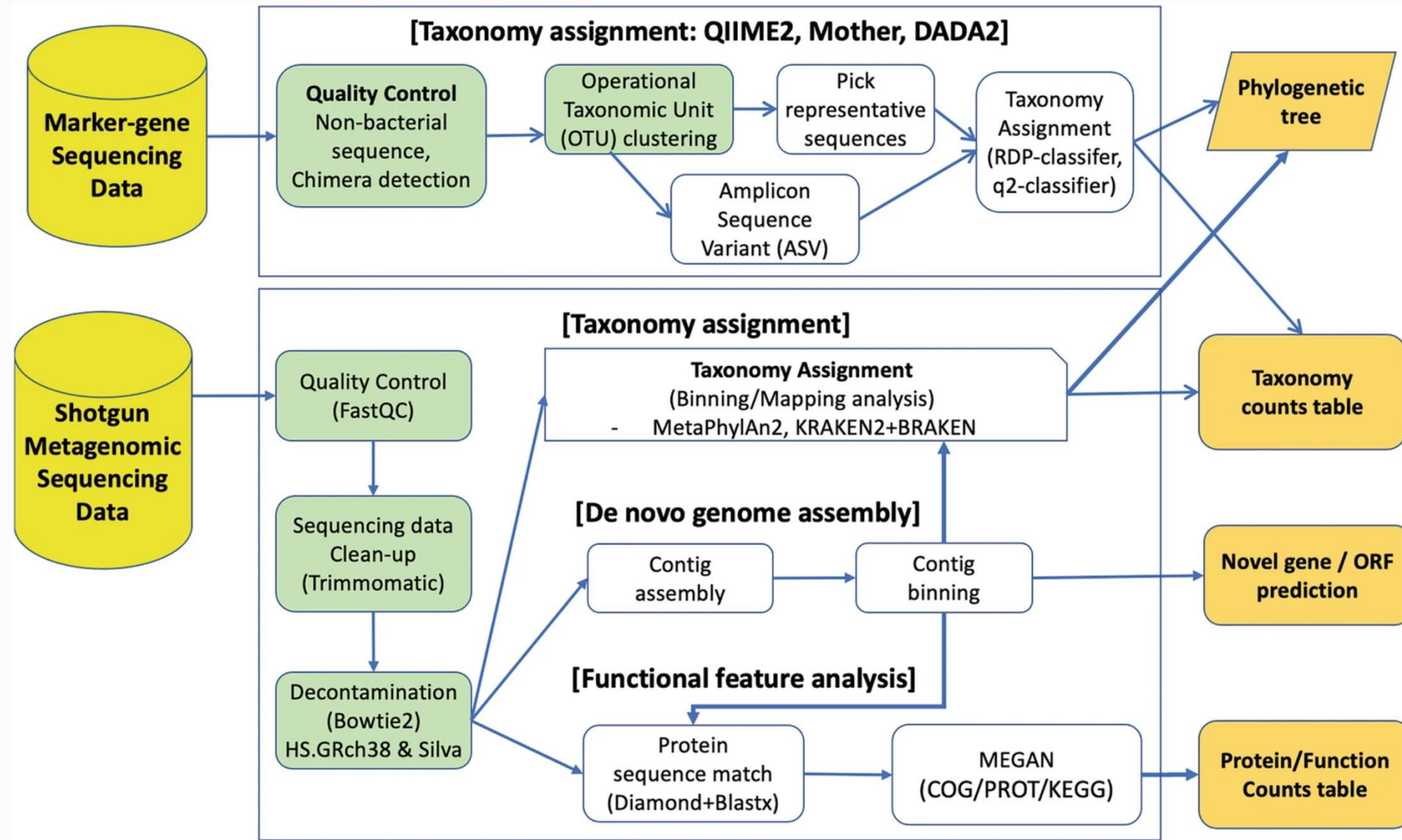


Dr. Brigida Rusconi
Department of
Pediatrics,
Washington
University School of
Medicine

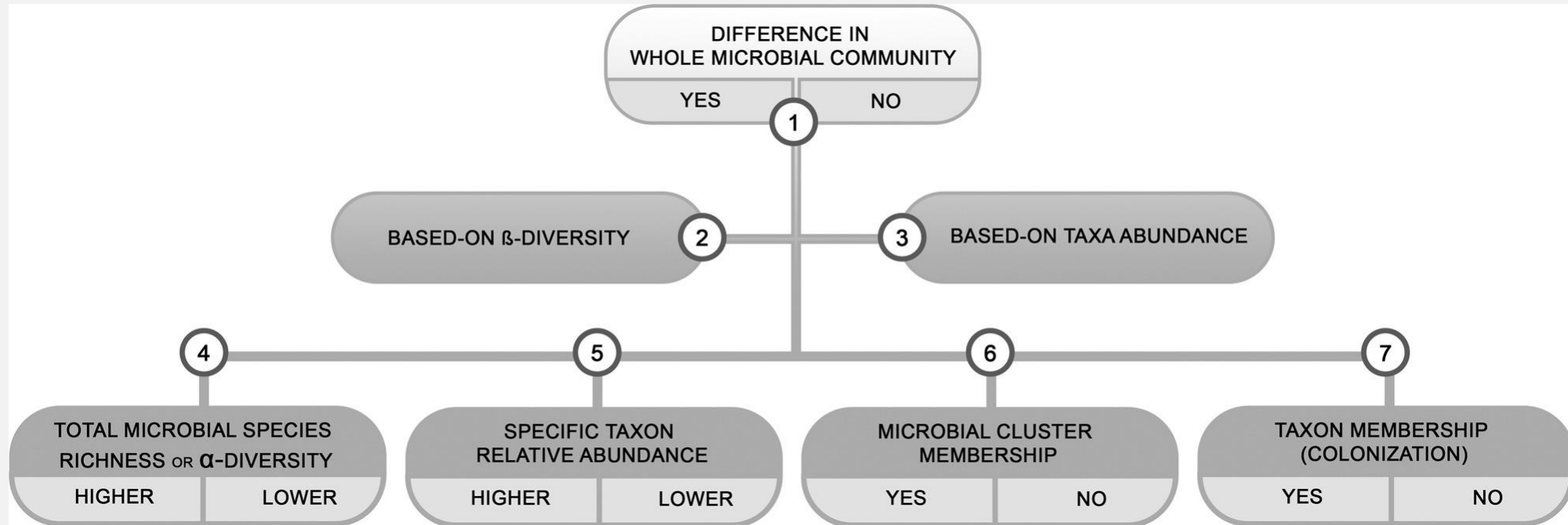
I 6S ANALYSIS

THE ROLE OF THE MICROBIOTA IN HEALTH AND DISEASE

- Many disease cannot be explained by genetics alone
 - Environmental variables that exacerbate or induce common human diseases
 - Sometimes the microbiota alone can be the trigger for disease
- Germ-free mice have an altered immune system and gut function
 - Colonization with gut microbes is required for tolerance against food antigens
 - Skews the immune system towards inflammation
- Perturbations in early life to the microbiota have life-long consequences
- Gut microbiota has systemic effects not just local



SAMPLE SIZE



Ferdous, 2022, Muc Imm

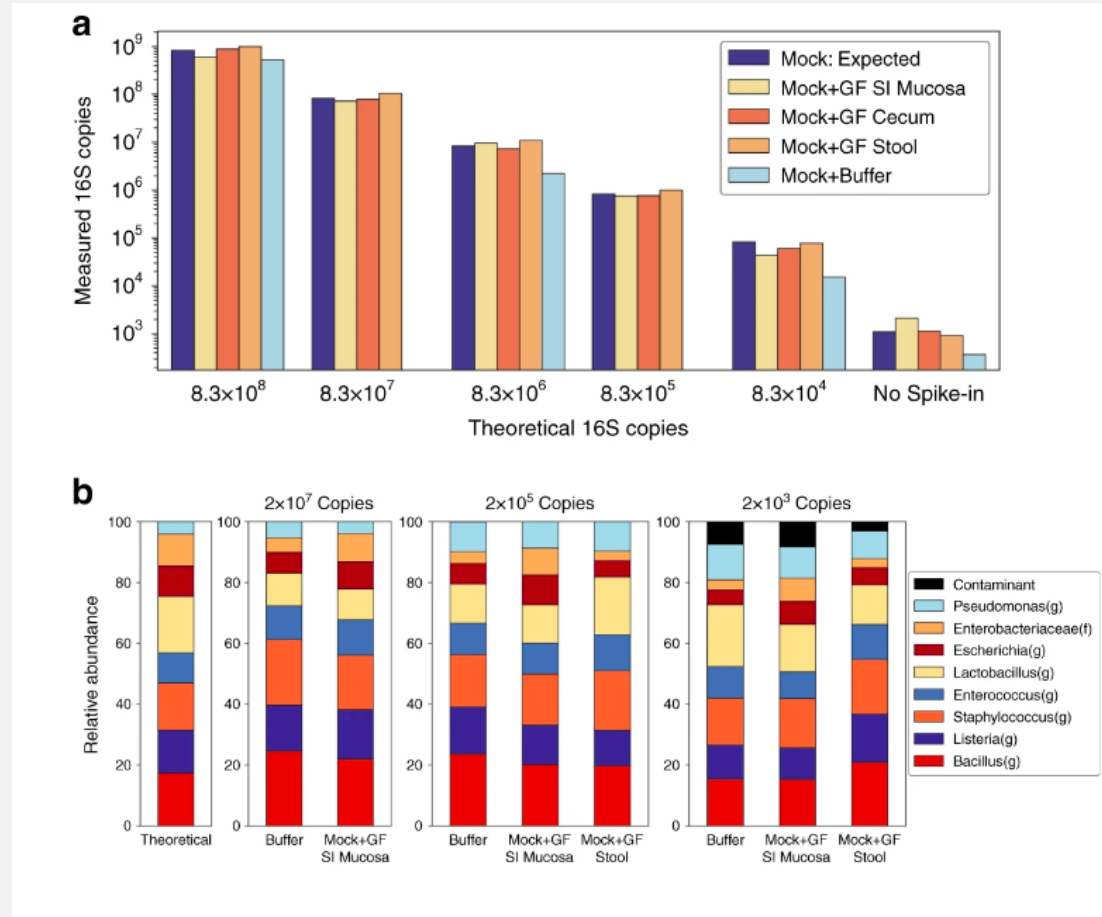
I6S OR SHOTGUN SEQUENCING?

- Depends on the budget and the scientific question
- Do you want to assess changes to community structure or potential function?
- Are you interested in rare or low abundance species?
- What is the source of your specimens (host or environment contaminants)?
- Do you need species or strain level resolution?
- If you are interested in function shotgun sequencing will provide better resolution
 - For functional information you will need higher read depth
 - Less samples per run
- Intermediate approach is to do shallow shotgun sequencing (500k/sample)
 - Increased taxonomic resolution
 - Computationally more demanding
 - Shallow shotgun sequencing cannot be used for novel gene and genome assembly

LOW ABUNDANCE SAMPLES

- What can I do to improve analysis of low abundance samples?
 - Enrich target DNA by degrading free DNA prior to DNA extraction (Stinson et al, 2019, Lett Appl Microbiol., Li et al., 2017, Scientific Reports)
 - Include an extraction control to get background amplification
 - This can then be used to remove background with package “decontam” (Davis et al, 2018, Microbiome)
 - Novel approach to prepare the samples to reduce host DNA and increase microbial reads
 - Microbial-enrichment method (MEM), has been validated on a wide range of sample types, including saliva, stool, intestinal scrapings, and intestinal mucosal biopsies (Natalie J. Wu-Woods, Nature Methods, 2023)
 - Recommended kits that have low contamination of bacterial DNA
 - Qiagen Dneasy Power soil Pro PN or MagAttract Power soil pro DNA kit,
 - Qiagen sells Microbial Free Water

LOW ABUNDANCE AND HOST DNA



Limit of detection

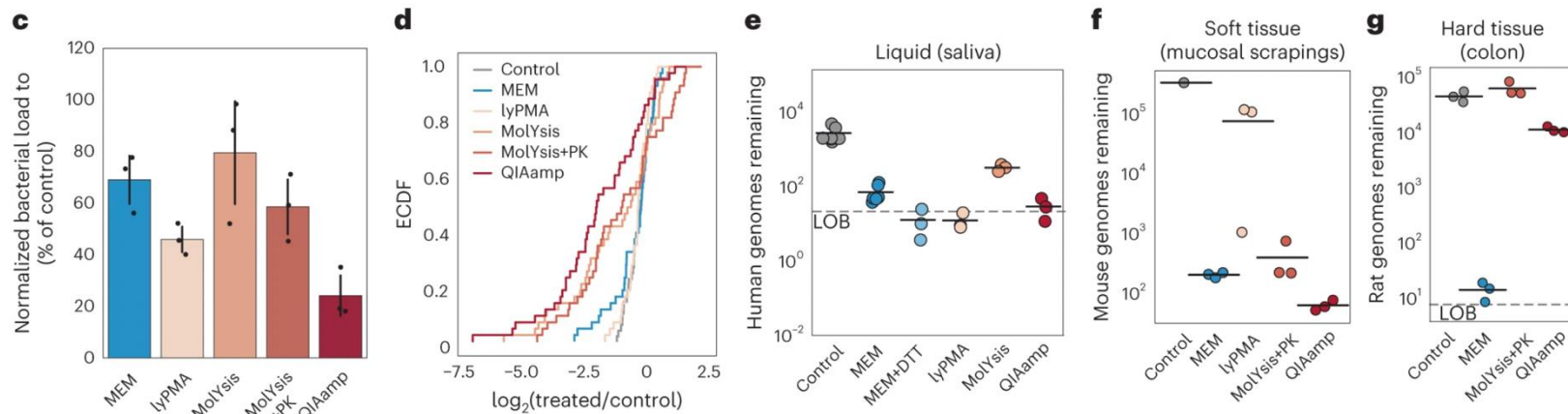
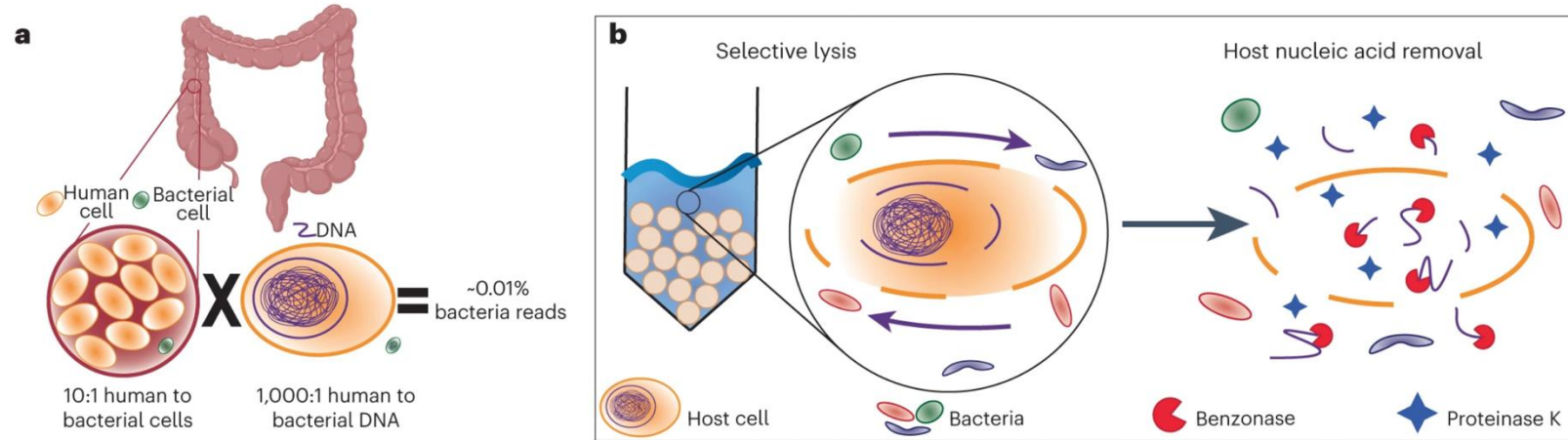
4.2×10^5 16S rRNA gene copies per gram for stool/cecum

1×10^7 16S rRNA gene copies per gram for mucosa

Usually extract
50-200mg stool/caecum
10mg of scraping

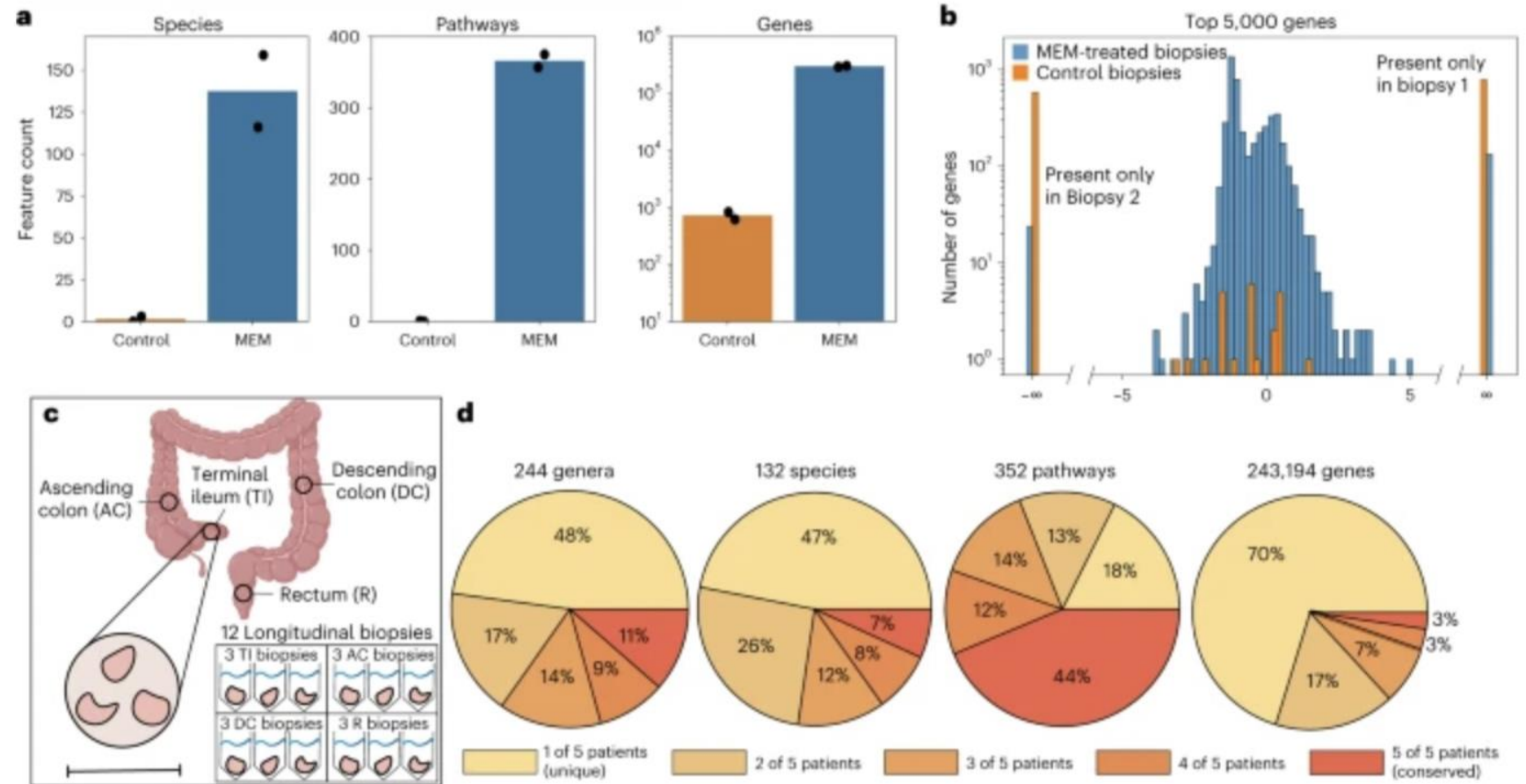
gDNA extraction columns cannot differentiate between host or bacterial DNA,
Columns can get saturated with host DNA which can lower the limit of detection

IMPROVED ISOLATION FROM SPECIMENS



INCREASED FUNCTIONAL RESOLUTION

Fig. 4: Shotgun sequencing of MEM-treated human intestinal biopsies.



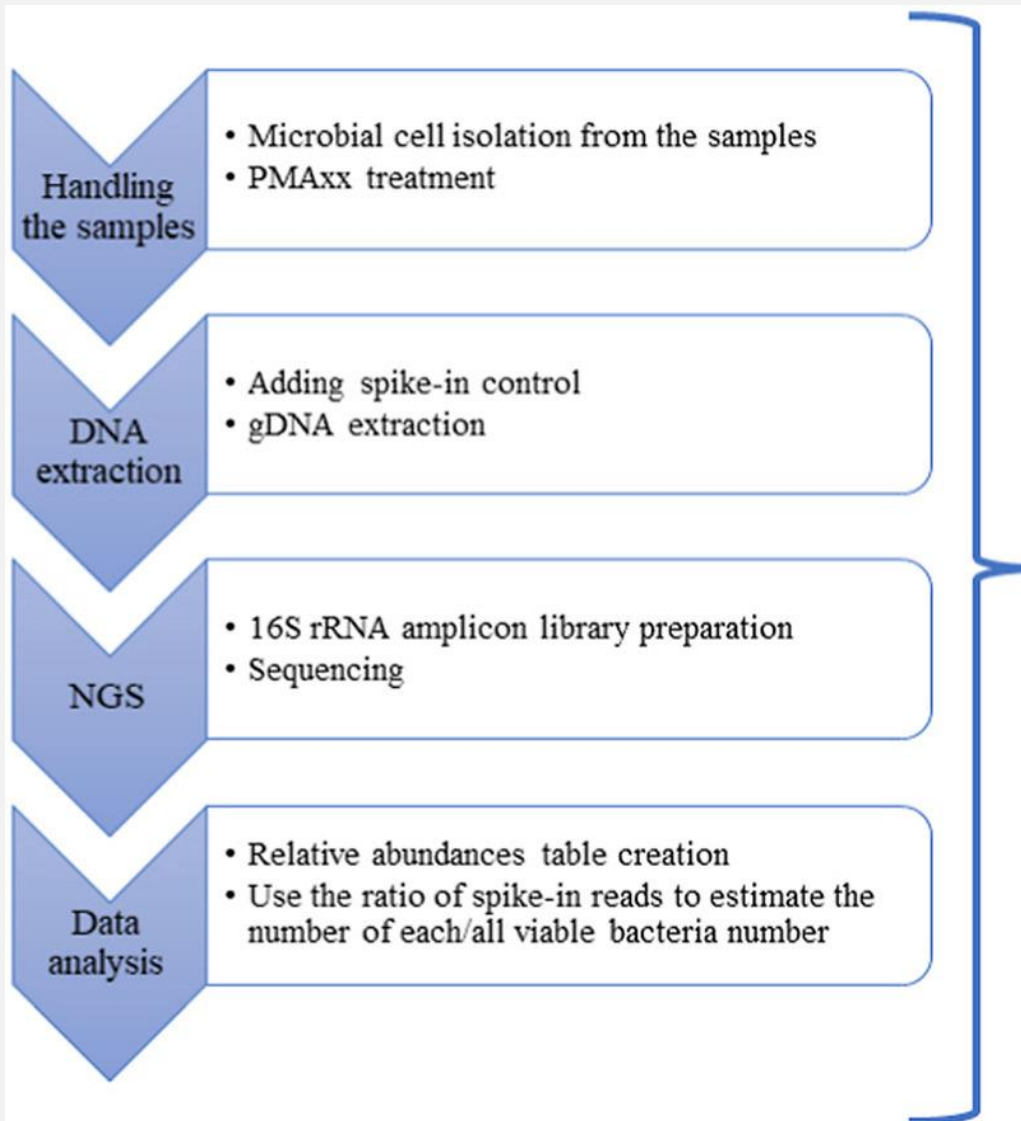
Classification of
taxa present
at 0.005%
compared to 10%
without

10^4 16S copies per
mg

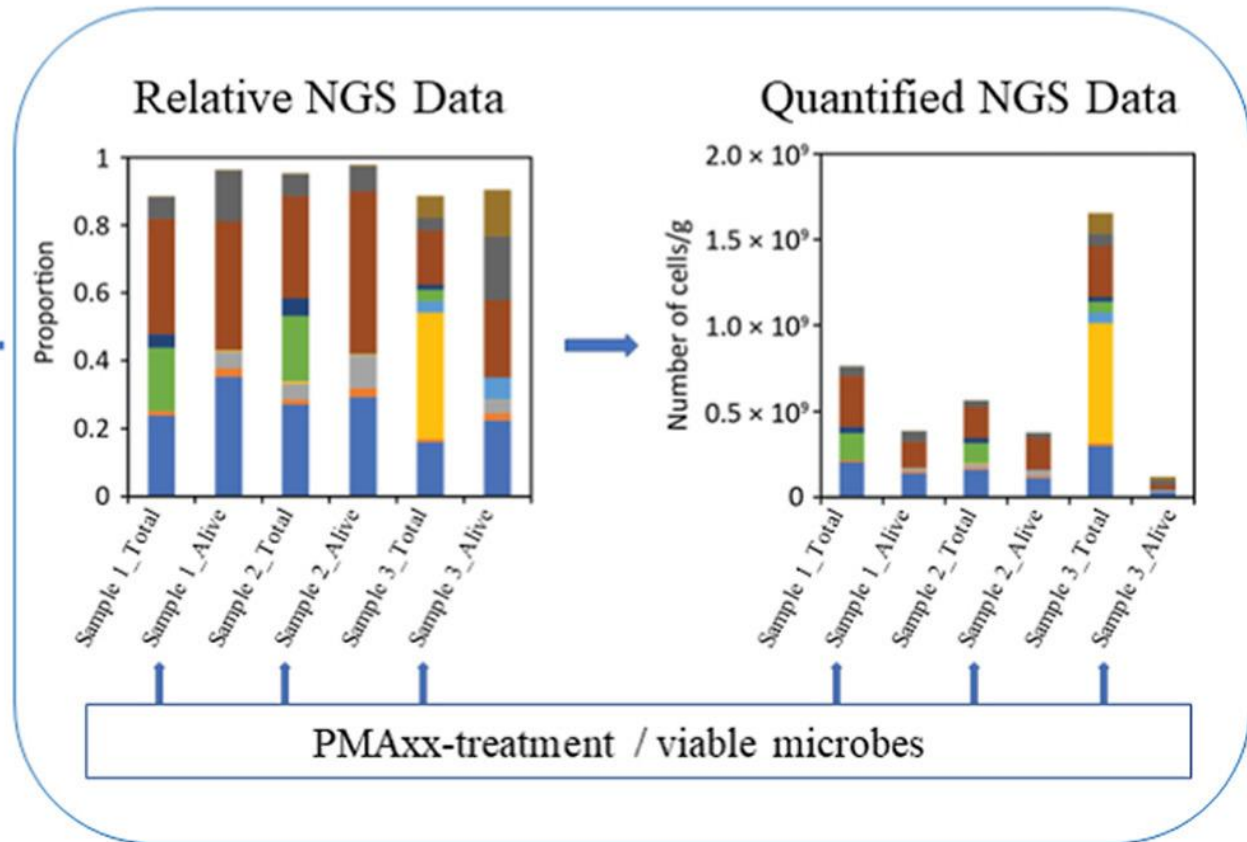
WHAT 16S CAN AND CANNOT DO

- Can
 - Rapid snapshot of bacterial community composition
 - Scalable for many samples
- Cannot
 - Full taxonomic resolution (species or strain)
 - Function of taxa present (strain differences, etc)
 - Absolute quantification
 - Only association without follow-up validation in animal studies

SPIKE-IN FOR ABSOLUTE 16S QUANTIFICATION



ZymoBIOMICS Spike-in Control



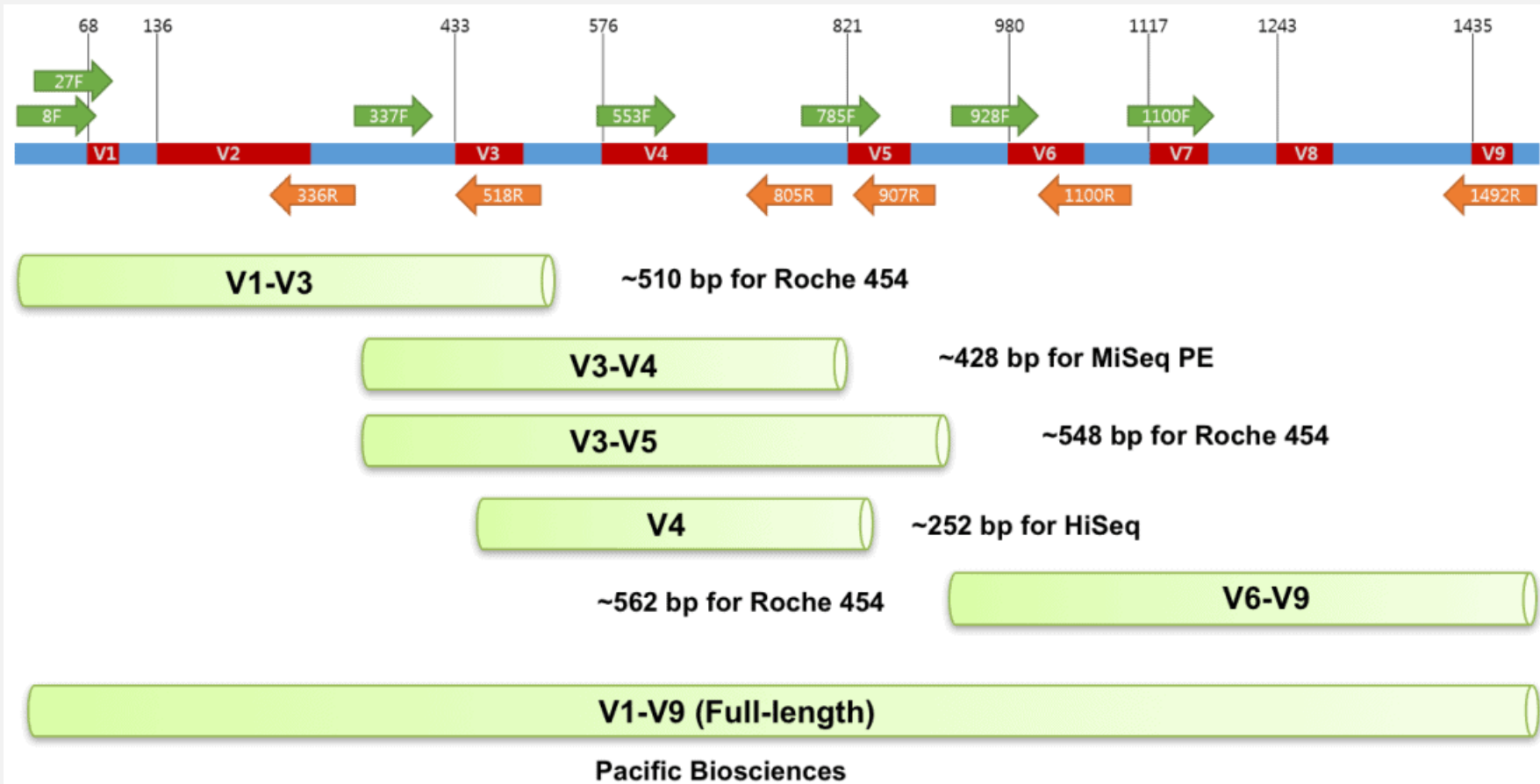
STRENGTHS OF 16S SEQUENCING

- Cheap and rapid data generation
- Can accommodate many samples on the same run
- Ideal to study conditions that change bacterial community patterns
- Multiple detailed tutorials and tools available to analyze the data
- Can compare datasets to previous publications (if you reanalyze the data with same parameters and if it's the same sequencing technology)
- Not as computationally intense as shotgun sequencing
- Low biomass ok (e.g. swabs, tissue, etc)

COMMON ISSUES WITH 16S

- Imperfect recall (not all sequences or taxa are detected)
 - Bias in extraction (universal)
 - Inefficiency of universal primers to hybridize all the templates
 - V4 sub-region has complete overlap of paired-end sequences
- 16S rRNA genes that do not differ in their amplified sequence cannot be resolved
 - Can only resolve on the family or genus level (Bukin et al, 2019, Scientific Data)
 - *Enterobacteriaceae* and the *Clostridiales* order are known to be poorly resolved using V4 amplicons
 - V2-V3 is better at resolving beyond the Genus level

WHICH VARIABLE REGION SHOULD I SEQUENCE?



WHICH VARIABLE REGION SHOULD I SEQUENCE?

- Depends on the source of your sample
- Most cohort studies of the past have used V4 for profiling
- 515f/806r primer pair (V4) originally described by Caporaso et al, 2011 has been used in almost 4000 publications

DO IT IN HOUSE OR SEND OUT?

- Budget
- Time-frame
- Sample #
- How complex is the analysis?
- Services available at WashU:
 - GTAC
 - Miseq
 - Shotgun
 - I6S full length with PacBio

APPROACHES TO ANALYZE 16S

- Generation of amplicon sequence variants (ASVs) using error-corrected reads (DADA2, Deblur)
- Direct taxonomic classification of raw reads (Kraken, Bracken)
 - Classifies sequencing reads into taxonomic bins
 - Bracken extension now also offers read count
 - Read that has identical matches to two species will be classified at the genus level
 - Multiple reads that can only be assigned at the genus level will be combined
- Alignment-free algorithm makes it really fast (Lu et al, 2020, Microbiome)

The *Lactobacillus* taxonomy change has arrived! What do you need to know?

🕒 April 21, 2020 👤 Nina Vinot and Marco Pane

Editor's Choice, Pharma & Human Health

Clostridium cluster IV

Clostridium cluster IV is now classified as several genera, but not all genera, in the family *Ruminococcaceae*.

Clostridium cluster XIVa

Clostridium cluster XIVa is now classified as several genera, but not all genera, in the family *Lachnospiraceae*.

[Bacteria](#); [Firmicutes](#); [Clostridia](#); [Clostridiales](#); [Lachnospiraceae](#)

16S DATA STRUCTURE

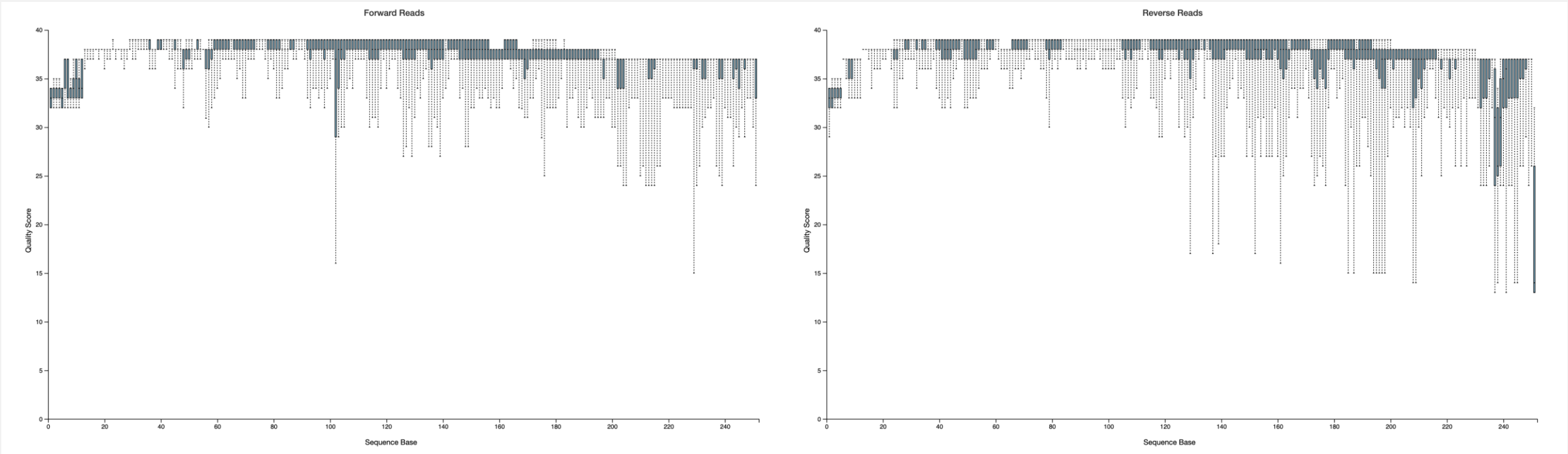
- More features (ASV) than samples (high dimensional)
- Many features with 0 (sparse)
- Number of reads between samples can vary
- Compositional (we only get a snapshot of the whole community)
 - Unless we use spike ins

PRE-REQUISITES FOR ANALYSIS

- Samples have been demultiplexed, i.e. split into individual per-sample fastq files
- Non-biological nucleotides have been removed, e.g. primers, adapters, linkers, etc
- If paired-end sequencing data, the forward and reverse fastq files contain reads in matched order
- Most sequencing services and cores will provide data in this format
- Have access to RIS or own machine with enough memory to run either RStudio or Qiime2 docker image

QIIME 2

- Best maintained repository of tools for microbiome data
- Can handle 16S amplicons and shotgun data
- Can do longitudinal analysis
- Docker image maintained by developer
- Has very active community that helps with questions
- Detailed tutorials
- Not as custom as having your own scripts
- Has a tracking system so you can reconstruct the commands run on the data (useful to track filtering, cutoffs, etc)
- Command-line based and requires a website to visualize the data
 - Stores data in “artefacts” which are just zipped text files
 - These files can then be further processed for visualization
 - Can also be run from python or R if you installed it locally



sample-id	input	filtered	percentage of input passed filter	denoised	merged	percentage of input merged	non-chimeric	percentage of input non-chimeric
#q2:types	numeric	numeric	numeric	numeric	numeric	numeric	numeric	numeric
016287d9	413	359	86.92	303	293	70.94	293	70.94
027ff912	654	596	91.13	521	458	70.03	458	70.03
05751894	658	569	86.47	521	514	78.12	514	78.12
059c4922	914	839	91.79	778	729	79.76	729	79.76
073e29ef	286	267	93.36	120	88	30.77	88	30.77
0948e1f5	903	827	91.58	749	703	77.85	703	77.85
09fe1	726	659	90.77	597	533	73.42	533	73.42

TAXONOMY

- Taxonomic assignment considerations
- ASVs will only get species level annotation if there is an exact 100% match to the database
- There are multiple species that have exact same variable regions (e.g. Escherichia coli and Shigella)
 - Change default parameters to allow multiple species names if you want to capture all matching species names (see option in Rmd file)
- Which database to compare to
 - Silva (most comprehensive database has a lot of environmental sequences)
 - RDP (mainly based on human microbiome data)
 - Greengenes (highest taxonomic accuracy but outdated naming 2012-2013)
- QIIME2 offers multiple trained taxonomic classifiers that you can use otherwise you can train your own classifier if you have a curated list of sequences with specific annotations that you want to use as a reference
 - [classifier](#)
- Before starting a new project or before publication I recommend checking if there is an updated in the reference database to have the most up-to-date nomenclature

TAXONOMY DATABASES

! Security risk

Pre-trained classifiers that can be used with q2-feature-classifier currently present a security risk.

This is due to the fact that unlike most QIIME 2 artifacts that only contain data, **classifiers contain [serialized Python objects](#) that contain a program that will run on your computer.** We have no way of vetting these programs and making sure they are safe.

If using a pre-trained classifier, such as the ones provided here, you should trust the person who trained the classifier and the person who provided you with it.

After downloading a classifier from this page, it is a very good idea to calculate the [sha256 sum](#) of the classifier using the command `shasum -a 256 <path-to-classifier>`. If the output does not match the SHA256 value associated with the classifier on this page, do not use the classifier and instead re-download it from this page. If this persists after you try downloading multiple times, please let us know on the [QIIME 2 Forum](#).

The safest thing to do, if you're concerned about using these classifiers, is to [train your own classifiers with QIIME 2](#) rather than download existing ones.

We hope to [address this security risk](#) in a future version of q2-feature-classifier. If you have relevant expertise and would like to help with this, we would welcome that!

TAXONOMY

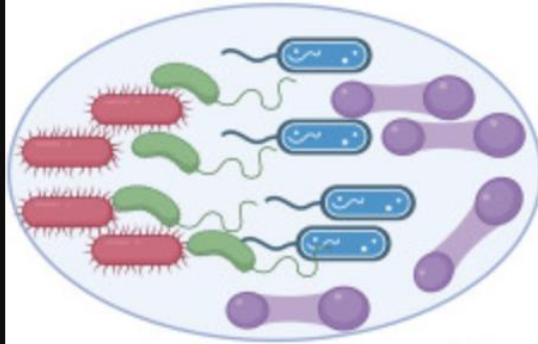
Sequence Table

To BLAST a sequence against the NCBI nt database, click the sequence and then click the *View report* button on the resulting page.

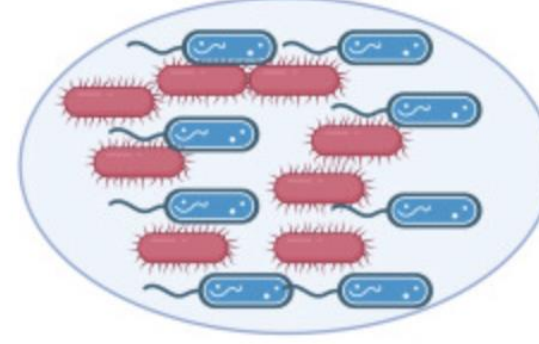
Download your sequences as a raw FASTA file

Click on a Column header to sort the table.

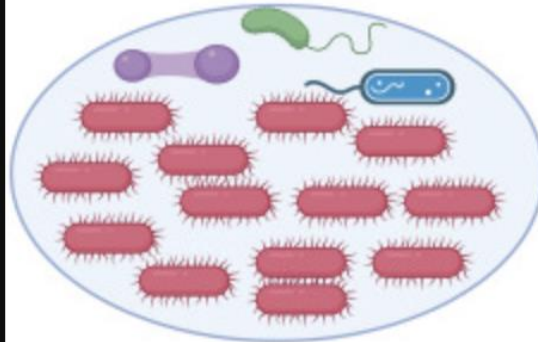
ALPHA DIVERSITY MEASURES



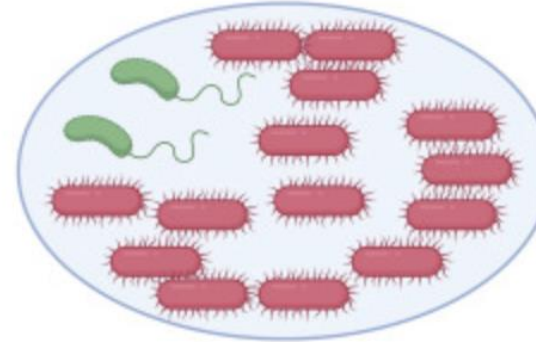
A. High richness and diversity (alpha diversity - high)



B. Low richness and diversity (alpha diversity - low to moderate depending on metric used)



C. High richness, low diversity (alpha diversity - depends on metric used*)



D. Low richness and diversity (alpha diversity - low)

Integrating sequence composition information into microbial diversity analyses with k-mer frequency counting

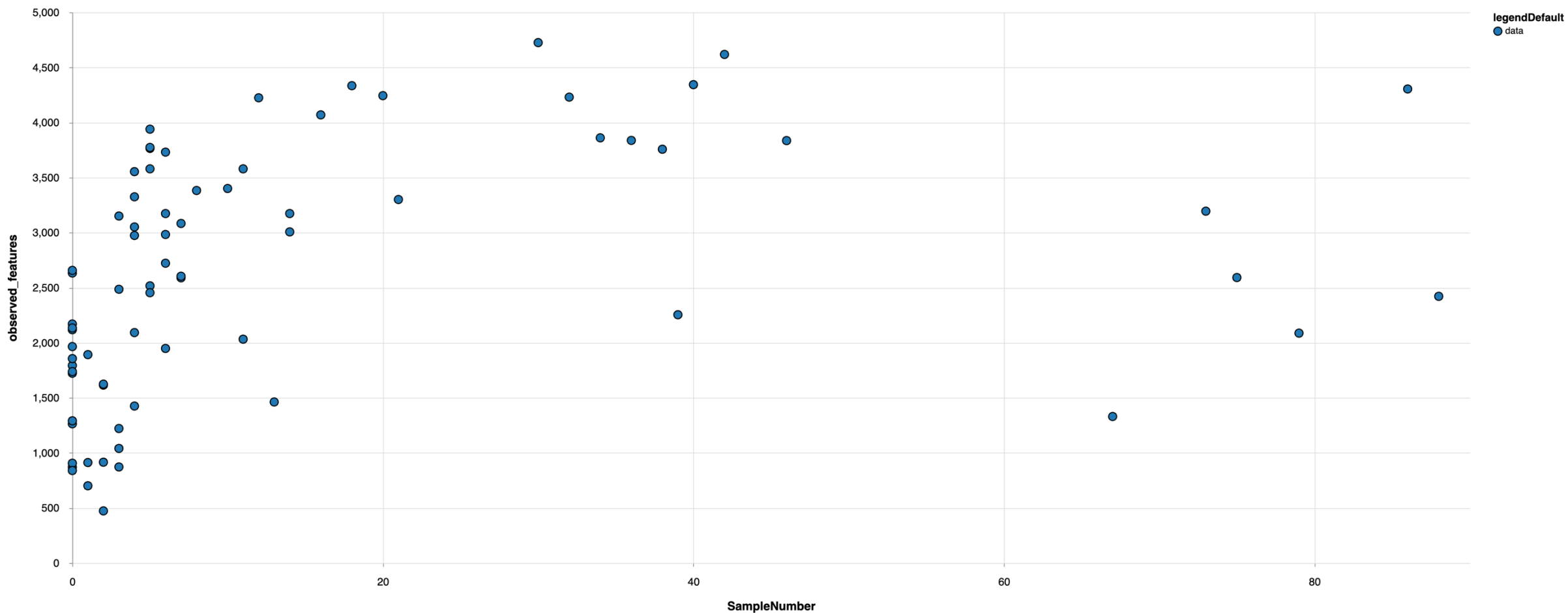
Nicholas A. Bokulich 

Department of Health Sciences and Technology, ETH Zurich, Zürich, Switzerland

- enabling large-scale reference-free profiling of microbiomes
- Does not require a phylogenetic tree generation (time consuming step)
- Has good correlation to classical alpha and beta diversity approaches
- Can also be used for fungi ITS sequences

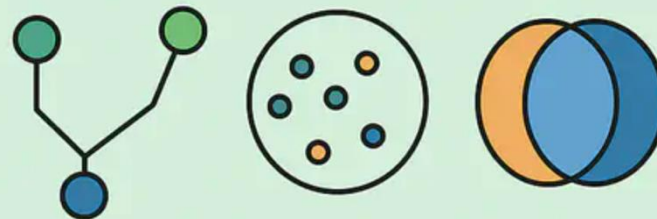
ALPHA DIVERSITY

Metric	Description	Strengths
Species Richness(Chao I,ACE, Good's coverage)	Counts the number of species present	Simple and intuitive
Shannon Index	Considers abundance and evenness	Reflects overall community structure
Simpson Index	Emphasizes dominant species	Useful for comparing communities
Pielou Evenness	Measure of community evenness	Reflects if there is a dominant species



BETA DIVERSITY

Metric	Description	Strengths
Jaccard Distance	Qualitative of community dissimilarity	Reflects presence absence without taking quantity into consideration (transfer of bacteria)
Bray-Curtis Distance	Quantitative of community dissimilarity	Reflects how similar each sample community is from another
Unifrac	Community dissimilarity based on phylogenetic relationship	Takes into account phylogenetic proximity



UniFrac Bray-Curtis Jaccard

title

xField

braycurtis Axis 1 (25%)

yField

braycurtis Axis 2 (10%)

markSize

150

includeZero

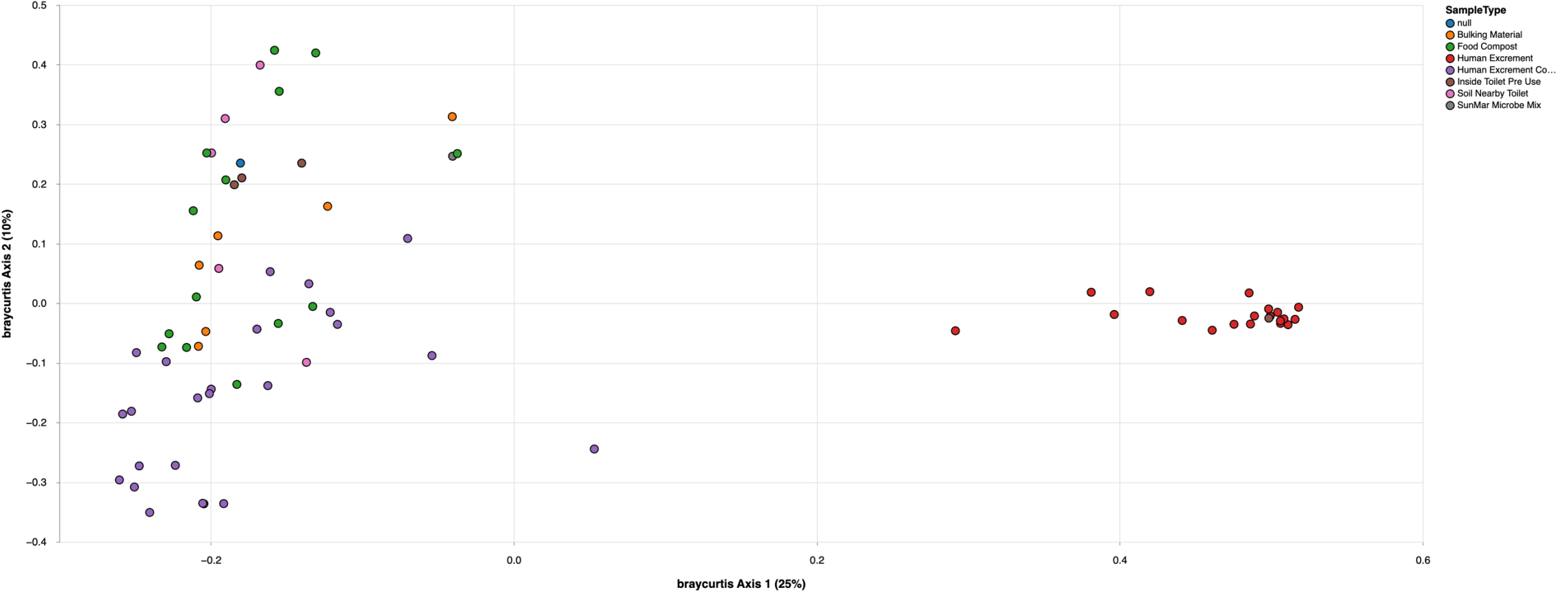
☒

colorBy

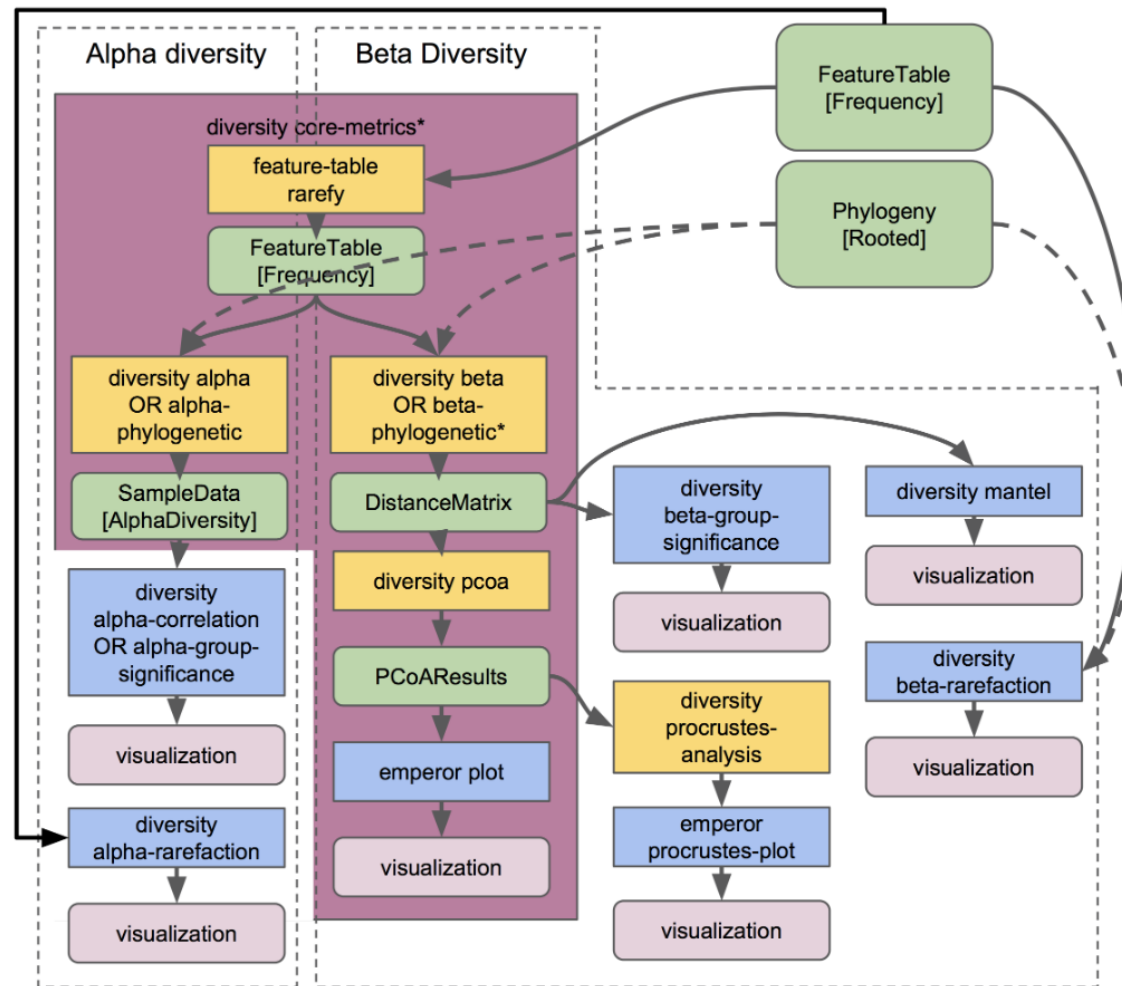
SampleType

colorPalette

category10



QIIME2 DIVERSITY OPTIONS



ZERO-INFLATION

- Zero inflated models should be considered for differential abundance:
 - If you compare samples from very different sourcing sites
 - If you do a treatment or intervention that would cause the depletion of given taxa
- This is important if you think that one of the groups will lack taxa not because you didn't sequence deeply enough, but because they are truly gone
- More information on <https://www.nicholas-ollberding.com/post/observation-weights-for-differential-abundance-of-zero-inflated-microbiome-data-with-deseq2/>
- Calgaro et al, 2020, Genome Biology

- Sample
- You need human
- It's best to be consistent
- When you have a sample with a

Differential abundance testing is easy to get wrong! 💀

Accurately identifying individual features that are differentially abundant across sample types in microbiome data is a challenging problem and an open area of research, particularly if you don't have an *a priori* hypothesis about which feature(s) are differentially abundant. A q-value that suggests that you've identified a feature that is differentially abundant across sample groups should be considered a hypothesis, not a conclusion, and you need new samples to test that new hypothesis.

In addition to the methods contained in the [composition plugin](#), new approaches for differential abundance testing are regularly introduced. It's worth assessing the current state of the field when performing differential abundance testing to see if there are new methods that might be useful for your data. If in doubt, consult a statistician.

sues

lly in

taxa

nfirm

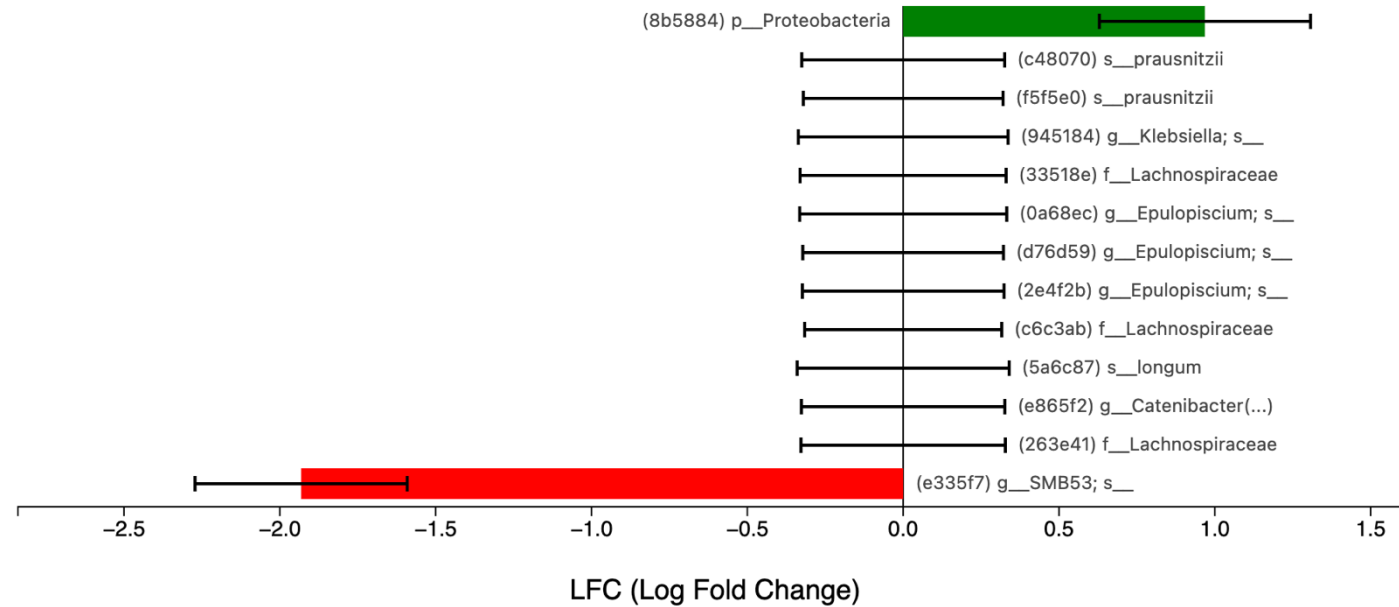
DIFFERENTIAL ABUNDANCE

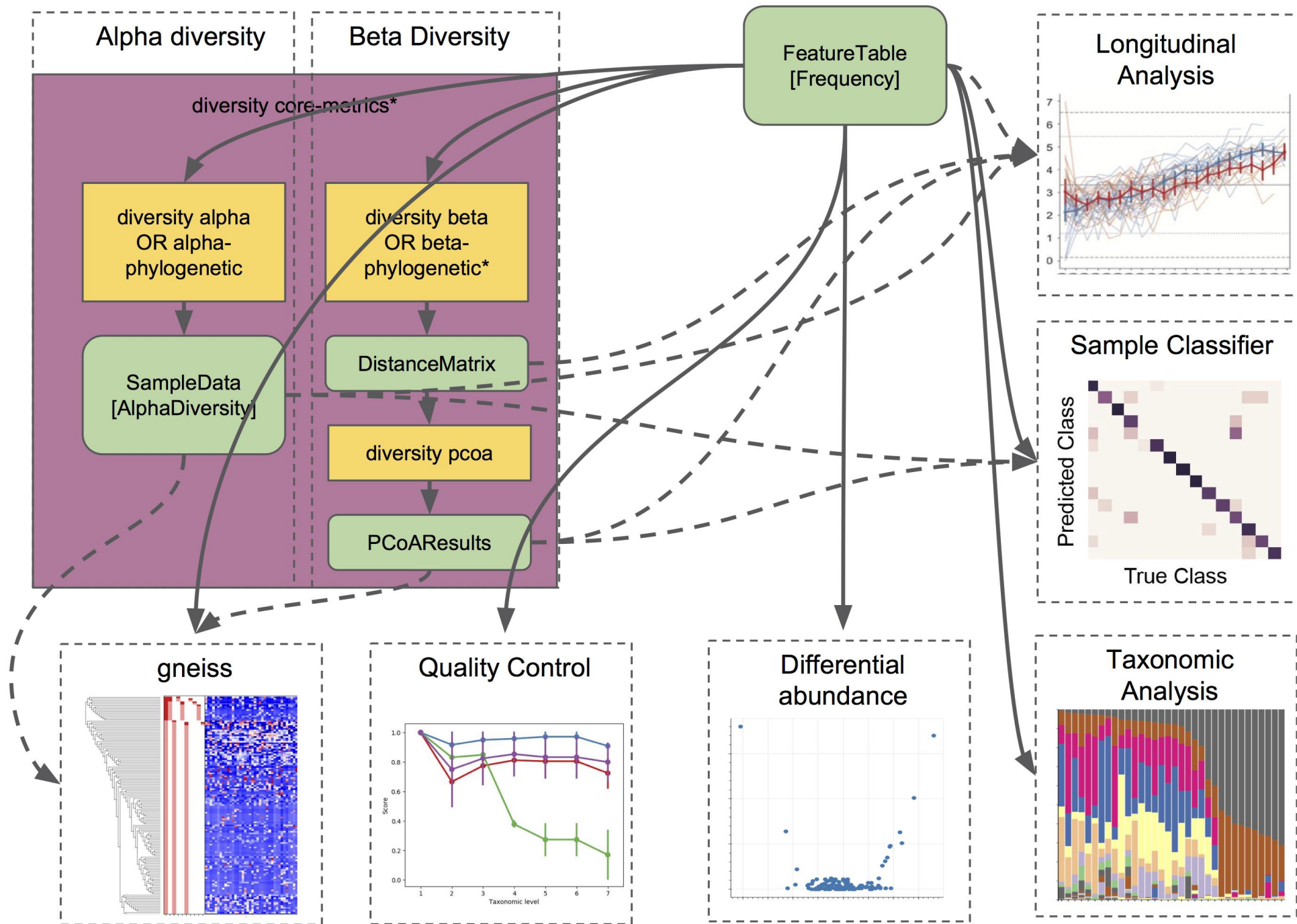
Method	Software	Approach	Type of filtered features	Absolute/relative differential abundance	Support to taxon bias correction
ALDEx2	R package ¹	Bayesian estimation of true relative abundance by Monte Carlo sampling. Centered log ratio (clr) transformed data are tested and mean p-values and q-values are provided after statistical test.	All zero features in both groups.	Relative abundances with respect to the geometric mean	No
ANCOM-II	R script ²	Test the difference between true absolute abundances using the log ratios of the observed absolute abundances (i.e. the count matrix). Each log ratio between the features of the subjects belonging to the same experimental group is described with an ANOVA model. The model's coefficients serve to perform the statistical test for each log-ratios.	• Features that have a percentage of zeros across all samples/ subjects greater than a predefined <i>zero_cut</i> threshold. • Structural zeros are considered as differentially abundant and removed from the dataset.	Absolute abundances	No
ANCOM-BC	R package ³	Estimate the unobservable sampling fraction c_j , correcting the bias introduced by the possible extreme variation between the subjects. A linear model with a sample-specific offset term estimated from the observed absolute abundances (i.e. the count matrix) is used to describe the log true absolute abundance of feature i in subject j .	• Features that have a percentage of zeros across all samples/ subjects greater than a predefined <i>zero_cut</i> threshold. • Structural zeros are considered as differentially abundant and removed from the dataset.	Absolute abundances	No
Corncob	R package ⁴	It estimates expected true relative abundances from the observed absolute abundances (i.e. the count matrix) fitting a beta-binomial model. The parameters are estimated through maximum likelihood and two different testing procedures are implemented: Likelihood-ratio (LRT) or Wald tests.	• All zero features in both groups. • Features for which the method fails to estimate the model parameters.	Relative abundances	No
DESeq2	R package ⁵	Negative binomial distribution is exploited to model observed absolute abundances. Relative log expression (RLE) is the default normalisation applied to observed absolute abundance (i.e. the count matrix). After model fitting, Wald test is used to evaluate taxa difference between groups.	Rare or outlier features identified using a procedure based on the Cook's distance.	Absolute abundances	Yes
eBay	R package ⁶	Empirical Bayesian estimation of mean posterior distribution of true relative abundance. Centered log ratio (clr) transformed proportions are tested to obtain the p-values and q-values.	None	Relative abundances with respect to the geometric mean	No
edgeR	R package ⁷	It is assumed that observed absolute abundances (i.e. the count matrix) follow a negative binomial distribution. Trimmed mean of M values (TMM) is the default normalisation applied to observed absolute abundance. After estimating model parameters, likelihood-ratio test (LRT) is used to test differentially abundant features.	None	Absolute abundances	Yes
MaAsLin2	R package ⁸	Log-transformed linear model on Total Sum Scaling (TSS)-normalised observed abundances.	Features with minimum prevalence at 10%.	Relative abundances	No
metagenomeSeq	R package ⁹	Observed absolute abundances (i.e. the count matrix) are modelled through zero-inflated Log-Normal mixture model using the built-in normalisation cumulative sum scaling (CSS). The estimated FC parameter is involved in the formulation of the statistical test.	None	Absolute abundances	No

Cappellato M, Baruzzo G, Di Camillo B (2022) Investigating differential abundance methods in microbiome data: A benchmark study. PLOS Computational Biology 18(9): e1010467.
<https://doi.org/10.1371/journal.pcbi.1010467>
<https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1010467>

ANCOM-BC2

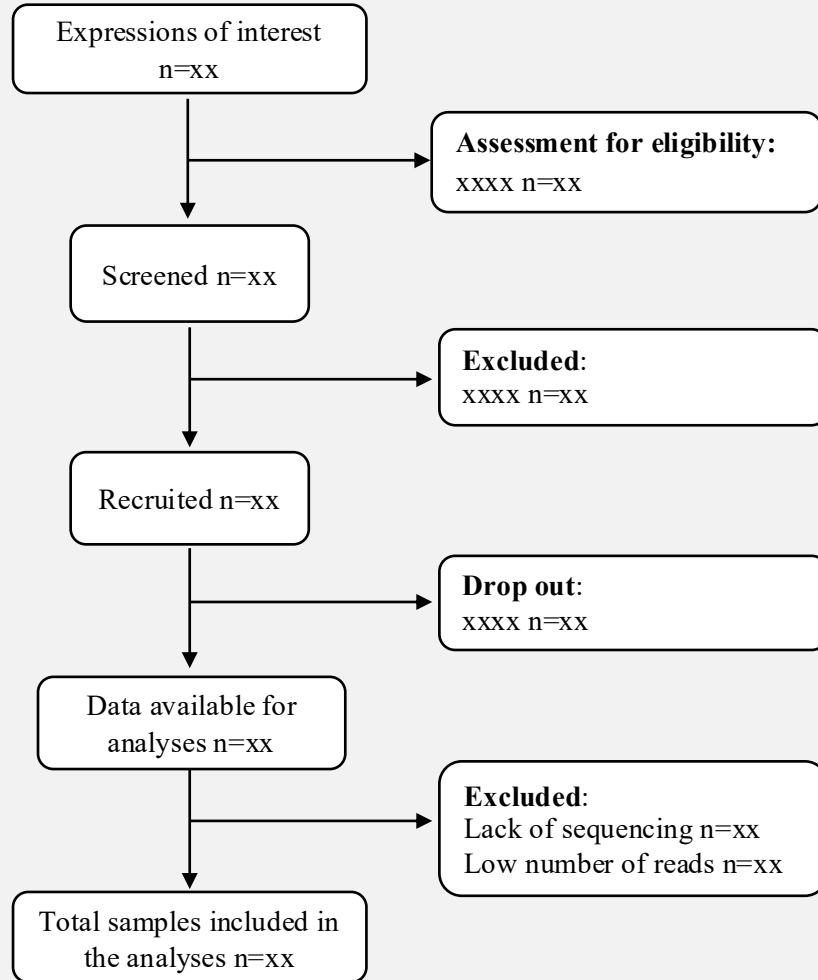
Reference Level: Human Excrement Compost





STORMS analytic sample size flowcharts (item 3.6)

Observational



Experimental

